

Advisor Email Record

Emails That Shaped the Research Direction

May–July 2026

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Faculty mentor: Woody Zhu

Graduate mentor: Wenbin Zhou

Document compiled 26 July 2026

Source PDF: `Advisor_Email_Record_Research_Direction.pdf`

Primary documentary record. Advisor-authored guidance is distinguished from student proposals and administrative context.

Purpose and Provenance

This document contains the email threads supplied by Nick Mino that materially shaped, clarified, or documented the evolution of the research project. The emails are ordered chronologically. Each entry begins with an annotation explaining its authority, significance, and relationship to the developing research direction. Routine scheduling material is omitted unless it establishes the occurrence or context of an important research meeting.

How to Read This Record

- Advisor-authored guidance has the highest evidentiary weight.
- Student-authored descriptions document the proposal presented to the advisors, but do not by themselves establish advisor approval.
- Later advisor formulations may supersede earlier student formulations.
- The June 11 email thread establishes the location and occurrence of an in-person meeting; substantive content from that meeting comes from contemporaneous notes and belongs in the main research packet, not in this email record.

Executive Summary

The email record shows a clear progression. Professor Zhu first directed Nick toward the group’s uncertainty-quantification and decision-focused papers and connected him with Wenbin Zhou for closer technical guidance. By late June, Nick proposed studying stabilized selection algorithms through their primitive operations and recurring stabilization mechanisms. After the June 29 meeting, Wenbin reported that he had rewritten the problem setup in Overleaf so that it aligned more closely

with what the advisors intended, and instructed Nick to begin developing methodologies for that formulation. A June 12 administrative exchange also records Wenbin’s recommendation that the project description explicitly use the term “post-hoc inference.”

Central evidentiary conclusion. The current formulation should be grounded in Wenbin’s July 2 advisor-authored problem setup and the June 29 discussion, while Nick’s June 25 primitive-operation framework should be treated as a student proposal that motivated that discussion rather than as the final approved formulation.

Chronological Timeline

Date	Authority	Importance	Research significance
May 11–14	Student request / advisor direction	High	Professor Zhu directs Nick to prioritize the previously supplied papers and establishes a recurring meeting structure.
May 18–20	Advisor direction	High	Professor Zhu connects Nick with Wenbin to clarify and outline the research objectives based on the proposal.
June 11	Meeting logistics	Contextual	Confirms the in-person meeting with Wenbin in the Teresa Heinz collaboration space.
June 12	Advisor terminology guidance	High	Records that Wenbin requested the phrase “post-hoc inference” in the formal project description.
June 25–July 3	Student proposal followed by advisor reformulation	Critical	Nick proposes a primitive-operation framework; after the June 29 meeting, Wenbin drafts a problem setup that “aligns closer to what we intended.”

1 Initial Reading Priorities and Working Structure

EXTREMELY IMPORTANT SOURCE — Email thread 1 (May 11–14)

Date: May 11–14, 2026

Participants: Nick Mino; Professor Woody Zhu

Subject / thread: Starting the research work effectively

Authority: Advisor direction responding to student request

Importance: High

Status: Foundational background direction; later narrowed by subsequent guidance

Why this matters. This thread establishes the first concrete working instructions after the project began. Nick asks which materials and concepts to prioritize, and Professor Zhu directs him to work through the papers already supplied before reconnecting. Professor Zhu also indicates that a PhD student will provide closer guidance as the project develops.

Key contributions.

- Establishes the initial literature-reading phase.
- Creates an expectation of recurring meetings.
- Foreshadows the later introduction of Wenbin as the primary technical guide.

Relationship to the research direction. This is background-setting rather than the final research formulation. It explains why the project initially centered on uncertainty quantification and the group's existing papers.

Full email thread

On Thu, May 14, 2026 at 12:45 AM, Woody Zhu <shixianz@andrew.cmu.edu> wrote:

Sure thing. We will organize meetings on a weekly basis. Sorry for my delayed response — I have been traveling lately.

In the meantime, please go through the papers I sent earlier, and let's reconnect next Wednesday once I return to Pittsburgh.

After you finish the reading, I will connect you with one of our PhD students, who can provide more guidance as you get further into the project.

Lastly, it is also perfectly fine to mention the research experience on LinkedIn.

Best,

Woody

On Mon, May 11, 2026 at 1:39 PM, Nick Mino <nmino@andrew.cmu.edu> wrote:

Hello Professor Zhu,

I wanted to ask what you would recommend I do first to start the research work effectively.

I have already started reading additional papers and trying to build a stronger general understanding of uncertainty quantification, but I want to make sure I am prioritizing the right background material and not moving in an unproductive direction. Are there specific papers, notes, codebases, or concepts you think I should focus on first?

I also wanted to ask what the expected structure will be for the research. Will there be in-person meetings, online meetings, asynchronous materials, or some combination of those? I am generally

available after 12:30 PM on weekdays and flexible outside of that, so I should be able to adjust to whatever meeting format works best.

Separately, I wanted to ask how I should describe the role publicly and professionally. Would it be appropriate for me to make a LinkedIn post mentioning that I am starting research work, or would you prefer that I wait? Also, for LinkedIn and my resume, what title would be most accurate: "Research Assistant," "Undergraduate Research Assistant," or something else?

Best,

Nick

2 Professor Zhu Connects Nick with Wenbin

EXTREMELY IMPORTANT SOURCE — Email thread 2 (May 18–20)

Date: May 18–20, 2026

Participants: Nick Mino; Professor Woody Zhu; Wenbin Zhou

Subject / thread: Paper priorities and research-objective clarification

Authority: Direct advisor instruction

Importance: High

Status: Active guidance; establishes Wenbin’s role

Why this matters. Professor Zhu explicitly asks Nick to work with Wenbin to clarify and outline the overall research objectives based on the proposal. This is the formal handoff from broad reading under Professor Zhu to closer technical development with Wenbin.

Key contributions.

- Wenbin is introduced as a third-year PhD student and co-author on several relevant papers.
- Nick is instructed to clarify and outline the research objectives before the next meeting with Professor Zhu.
- Nick identifies the specific papers he had been reading: inverse conformal risk control, conformalized decision risk assessment, Gen-DFL, and hierarchical probabilistic conformal prediction.

Relationship to the research direction. This thread explains why later messages from Wenbin carry substantial authority over the technical problem formulation. It also records the literature base from which the project initially developed.

Full email thread

On Wed, May 20, 2026 at 11:04 AM, Woody Zhu <shixianz@andrew.cmu.edu> wrote:
Thank you, Wenbin!
Best,
Woody

On Wed, May 20, 2026 at 7:26 AM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:
Thank you, Woody, for connecting us.
Nick has reached out to me in a separate email, and I will chat more with him offline.
Best,
Wenbin

On Tue, May 19, 2026 at 8:12 PM, Woody Zhu <shixianz@andrew.cmu.edu> wrote:
Hi Nick,
Sorry for the delayed response. I’ve been traveling recently and will be fully back on campus after June 1st. After that, we can resume meeting on a more regular weekly basis.
In the meantime, I’d like to connect you with Wenbin Zhou, who is a third-year PhD student in our group and also a co-author on several of these papers.
Please feel free to reach out to Wenbin if you have questions about the papers or related background material. I would also suggest working with him to further clarify and outline the overall

research objectives based on the proposal before we meet again. I think that would help make our future discussions more focused and productive.

Thanks,

Woody

On Mon, May 18, 2026 at 12:07 PM, Nick Mino <nmino@andrew.cmu.edu> wrote:

Hello Professor Zhu,

Sorry for my late response, and I apologize for not making more progress on the reading sooner.

I have continued working through the material we discussed. I focused most closely on your papers on inverse conformal risk control and conformalized decision risk assessment, along with some related papers.

In your last email, you mentioned the papers you had sent earlier. I may have missed something, but I could not find a separate list or attachments. I have mainly been reading the papers we had discussed previously and some of your related work, including Gen-DFL and Hierarchical Probabilistic Conformal Prediction for Distributed Energy Resources Adoption.

Could you clarify which specific papers I should focus on before we reconnect?

Best,

Nick

3 Evidence of the June 11 In-Person Meeting

Date: June 8–11, 2026

Participants: Nick Mino; Wenbin Zhou

Subject / thread: Meeting location in Hamburg Hall

Authority: Administrative corroboration

Importance: Contextual but important

Status: Establishes occurrence and location; not substantive research guidance

Why this matters. This short thread confirms that Nick and Wenbin met in person on June 11, 2026, in the Teresa Heinz collaboration space on the B-level of Hamburg Hall. The substantive research insight from this meeting is preserved separately in Nick’s contemporaneous notes, because no transcript is available.

Key contributions.

- Confirms the date and physical location of the meeting.
- Supports the provenance of the June 11 contemporaneous research note.
- Should be cross-referenced by the main research packet.

Relationship to the research direction. The thread itself contains no technical research content. Its value is evidentiary: it validates that the undocumented in-person meeting occurred.

Full email thread

On Thu, Jun 11, 2026 at 3:07 PM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:

Hey Nick,

I’m at the Teresa Heinz collab space.

Best,

Wenbin

On Thu, Jun 11, 2026 at 3:05 PM, Nick Mino <nmino@andrew.cmu.edu> wrote:

I apologize for not asking this sooner, but where in Hamburg b level are we meeting? I have been in the Teresa Heinz collab space.

On Tue, Jun 9, 2026 at 11:41 AM, Nick Mino <nmino@andrew.cmu.edu> wrote:

B-level Hamburg Hall works for me.

Best,

Nick

On Mon, Jun 8, 2026 at 1:57 PM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:

Thanks. How about the basement (B-level) of Hamburg Hall?

4 Required Terminology: “Post-Hoc Inference”

EXTREMELY IMPORTANT SOURCE — Email thread 4 (June 12 terminology)

Date: June 12, 2026

Participants: Nick Mino; Wenbin Zhou

Subject / thread: SURA project description and signed form

Authority: Advisor terminology guidance embedded in administrative exchange

Importance: High

Status: Active terminology requirement

Why this matters. Although this exchange primarily concerns a signed form, Nick explicitly states that he incorporated Wenbin’s suggestion to add the phrase “post-hoc inference” to the project description. This is direct evidence that the inferential objective was intended to be part of the formal research framing.

Key contributions.

- Records Wenbin’s requested terminology.
- Shows that the project was not merely about generic stability or robust optimization.
- Supports using post-hoc inference as a defining component of the current problem statement.

Relationship to the research direction. This terminology constraint remains consistent with the later optimization-stability formulation. It should be treated as an active requirement unless superseded by explicit advisor guidance.

Full email thread

On Fri, Jun 12, 2026 at 3:10 PM, Nick Mino <nmino@andrew.cmu.edu> wrote:

Hi Wenbin,

Thank you, this version works. I appreciate your help.

Best,

Nick

On Fri, Jun 12, 2026 at 3:01 PM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:

Here is the PDF version.

On Fri, Jun 12, 2026 at 2:59 PM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:

Let me know if this works. There was probably some formatting issue that caused the signature to be unable to display.

On Fri, Jun 12, 2026 at 2:56 PM, Nick Mino <nmino@andrew.cmu.edu> wrote:

Hi Wenbin,

Thank you for sending this back. I incorporated your suggestion and added “post-hoc inference” to the project description.

I noticed that the phone number and date were added, but the additional mentor signature line at the bottom is still blank. Would you be able to sign that line in the attached updated version?

Best,

Nick

5 Primitive-Operation Proposal and Advisor Reformulation

EXTREMELY IMPORTANT SOURCE — Email thread 5 (June 25–July 3) — critical

Date: June 25–July 3, 2026

Participants: Nick Mino; Professor Woody Zhu; Wenbin Zhou

Subject / thread: Research approach, June 29 meeting, and revised problem setup

Authority: Student proposal followed by advisor-authored reformulation

Importance: Critical

Status: July 2 formulation supersedes earlier student wording where inconsistent

Why this matters. This is the most consequential email chain in the supplied record. Nick proposes a framework based on stabilized algorithms, primitive operations, problem structure, and downstream inference goals. After the June 29 meeting, Wenbin states that he drafted a new problem setup in Overleaf that “aligns closer to what we intended” and authorizes Nick to begin brainstorming methodologies for that formulation.

Key contributions.

- Documents Nick’s primitive-operation and stabilization-pattern proposal before advisor review.
- Establishes that the June 29 meeting was used to evaluate the current approach and SURA proposal diagram.
- Contains Wenbin’s explicit statement that his revised setup is closer to the advisors’ intended problem.
- Moves the project from problem clarification into methodology development.

Relationship to the research direction. Nick’s June 25 message is evidence of the proposed architecture, not conclusive evidence of advisor approval. Wenbin’s July 2 Overleaf formulation has higher authority and should control the current problem statement. The attached files and the actual Overleaf text are therefore essential companion evidence.

Full email thread

On Fri, Jul 3, 2026 at 2:24 AM, Nick Mino <nmino@andrew.cmu.edu> wrote:

Hi Wenbin,

Thank you for drafting the problem setup. I will read through it carefully and start brainstorming possible methodologies to add to the draft. I will let you know if I have any questions.

For the recurring meeting, I do not think we have one set up yet. I am available any weekday before 10:30 AM or after 12:30 PM. A weekly meeting would work well for me.

Best,

Nick

On Thu, Jul 2, 2026 at 12:53 PM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:

Hi Nick,

Nice talking to you on Monday, and I am glad to see that we are making good progress.

I drafted a version of the problem setup in the Overleaf project, which aligns closer to what we intended. Please feel free to give it a read and let me know if you have any questions. If not, you

may begin brainstorming the methodologies we could use to approach this problem and write them down in the draft. We can discuss them next time we meet.

Woody — Please see attached for the two files that Nick presented on Monday.

By the way, do we have a weekly recurring meeting with Nick set up? I don't see any on my calendar.

Best,
Wenbin

On Fri, Jun 26, 2026 at 12:56 PM, Nick Mino <nmino@andrew.cmu.edu> wrote:

That works for me, thank you.

On Fri, Jun 26, 2026 at 11:56 AM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:

Works for me as well.

On Fri, Jun 26, 2026 at 10:45 AM, Woody Zhu <shixianz@andrew.cmu.edu> wrote:

Sorry I realized I have a meeting between 4–5. How about 5–5:30?

Woody

On Fri, Jun 26, 2026 at 9:04 AM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:

Works for me

On Fri, Jun 26, 2026 at 8:02 AM, Woody Zhu <shixianz@andrew.cmu.edu> wrote:

What about 4–4:30?

Woody

On Fri, Jun 26, 2026 at 6:37 AM, Wenbin Zhou <wenbinz2@andrew.cmu.edu> wrote:

Hi Woody,

12:30–1:00 is a bit tough for me. Would any time before 12 or after 1 work?

Best,
Wenbin

On Thu, Jun 25, 2026 at 10:09 PM, Woody Zhu <shixianz@andrew.cmu.edu> wrote:

Hi Nick,

Thanks for reaching out. How about next Monday 12:30–1?

Let me know if this works for you and Wenbin.

Best,
Woody

On Thu, Jun 25, 2026 at 4:19 PM, Nick Mino <nmino@andrew.cmu.edu> wrote:

Hello Professor Zhu and Wenbin,

I hope you are doing well. I wanted to ask whether we could schedule a meeting soon to discuss my current research approach and the SURF Week 3 proposal diagram.

My current approach is to focus on algorithmic stability for post hoc selection/inference by studying concrete examples of selection algorithms that have been ‘stabilized’. I have been gathering these examples so that I can compare the structure of the original selection algorithm with the stabilized version, identify the primitive operations involved, and look for recurring patterns in how stability is introduced. I am in the process of converting the examples I found into LaTeX. The goal is to begin work on a framework that analyzes a selection algorithm’s operations, the relevant data type or problem structure, and the downstream inference or evaluation goal to determine an appropriate formulation for adding algorithmic stability.

I wanted to check whether this is the right approach to continue pursuing.

For the Week 3 assignment, I also need to ask a few questions about the research question, novelty, significance, methods, and expected outcomes, so the next meeting would be useful for confirming those pieces.

I also wanted to ask what schedule would make sense for regular meetings going forward. Would you prefer weekly or biweekly meetings, and is there a recurring time that generally works for both of you?

I am available any day before 10:30 AM or after 12:30 PM, except Saturdays.

Best,

Nick

Concluding Synthesis: Evolution of Advisor Guidance

1. **Initial background phase.** Professor Zhu first directed Nick to study the group’s uncertainty-quantification and decision-focused papers. At this stage, the technical problem remained broad.
2. **Technical mentorship and objective clarification.** Professor Zhu then connected Nick with Wenbin and explicitly asked them to clarify and outline the research objectives based on the proposal. This established Wenbin as the primary source of detailed technical guidance.
3. **Post-hoc inference becomes explicit.** The June 12 exchange records Wenbin’s suggestion that the formal project description include the phrase “post-hoc inference.” This indicates that downstream inferential validity is a defining objective, not an optional application.
4. **Student-proposed primitive framework.** By June 25, Nick proposed studying concrete stabilized selection algorithms, decomposing them into primitive operations, and identifying recurring stabilization patterns. This is the clearest email statement of the architecture Nick brought to the advisors.
5. **Advisor reformulation controls.** After the June 29 meeting, Wenbin drafted a new problem setup that he said aligned more closely with what the advisors intended. That formulation supersedes earlier student wording where the two differ and should be the primary source for the current mathematical problem.
6. **Methodology phase begins.** Wenbin then instructed Nick to begin brainstorming and writing methodologies for the revised setup. The project had therefore moved from broad reading and problem identification into the development of candidate methods.

Central evidentiary conclusion. The project concerns algorithmic stability for optimization-based selection in support of post-hoc inference, with the final formulation governed by the advisor-authored July 2 problem setup. The primitive-operation framework is an important proposed methodology, but its status must be evaluated against that advisor-authored formulation and the June 29 meeting transcript.
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Missing Companion Evidence

- The actual July 2 Overleaf problem setup.
- The two files attached to Wenbin’s July 2 email.
- The June 29 meeting transcript.
- The June 11 contemporaneous notes describing the optimization-specific stability intuition.
- Any earlier June 5 email in which Wenbin directly connected algorithmic stability to replacing data splitting in CREME, if available.

End of record.